ADR Identification Basics

Task 1: Analysis of patient case scenarios to identify potential Adverse Drug Reactions

Prepared by: Dr. Zainab Umar (BDS)

Objective

To review patient case scenarios, study the symptoms, medication history and timeline, decide whether each reaction is drug-related, and identify the suspected drug.

What is an Adverse Drug Reaction?

An Adverse Drug Reaction (ADR) is a response to a medicine that is noxious and unintended, and which occurs at doses normally used in humans for prophylaxis, diagnosis or treatment (WHO definition).

Approach used for each case

Each case is assessed by asking four questions:

1. **Temporal relationship:** did the symptoms start after the medicine was started, within a reasonable time?
2. **Known reaction:** is this a recognised reaction to that medicine?
3. **Alternative causes:** could the disease itself or another medicine explain it?
4. **Dechallenge / rechallenge:** did the reaction improve when the medicine was stopped, and did it return if the medicine was given again?

The answers are then matched to the WHO-UMC causality categories:

Category	Meaning
Certain	Plausible timing, cannot be explained by disease or other drugs, improves on withdrawal, and returns on rechallenge.
Probable / Likely	Reasonable timing, unlikely to be due to disease or other drugs, improves on withdrawal. Rechallenge not required.
Possible	Reasonable timing, but could also be explained by disease or other drugs. Withdrawal information lacking or unclear.
Unlikely	Timing makes a relationship improbable, and disease or other drugs give a plausible explanation.
Conditional / Unclassified	More data needed for a proper assessment.
Unassessable / Unclassifiable	Information is insufficient or contradictory and cannot be verified.

Reactions are also classified by type: **Type A (Augmented)** reactions are related to the pharmacological action of the drug and are predictable; **Type B (Bizarre)** reactions are unpredictable and not related to the usual pharmacological action, such as allergy.

Case 1: Skin rash after an antibiotic

Patient	Female, 45 years
Presenting complaint	Itchy red rash over the trunk and arms, with nausea and dizziness
Medical history	Treated for a dental infection. No known chronic illness. No previously known drug allergy.

Medication history

Medicine	Dose	Indication	Started
Amoxicillin	500 mg three times daily	Dental infection	Day 1
Paracetamol	500 mg when needed	Pain	Day 1 (used many times before without any problem)

Timeline

When	What happened
Day 1	Amoxicillin and paracetamol started.
Day 3	Itchy maculopapular rash appears on trunk and arms, with nausea and dizziness.
Day 4	Symptoms persist. Amoxicillin stopped; antihistamine given.
Day 8	Rash and other symptoms fully resolved.

Analysis

Question	Finding
Temporal relationship	Yes. Symptoms began 2 days after amoxicillin was started.
Known reaction	Yes. Skin rash is a well-recognised reaction to penicillins.
Alternative causes	Unlikely. Paracetamol was tolerated many times before; a dental infection does not explain a generalised itchy rash.
Dechallenge	Positive. The rash resolved after amoxicillin was stopped.
Rechallenge	Not done (not advisable in a suspected allergy).

Conclusion

Drug-related: Yes
Suspected drug: Amoxicillin
Causality (WHO-UMC): Probable / Likely
Type of reaction: Type B (hypersensitivity)
Seriousness: Non-serious
Action: Stop the drug, record penicillin allergy in the patient's notes, and advise the patient to avoid penicillins in future.

Case 2: Persistent dry cough after a new blood pressure medicine

Patient	Male, 58 years
Presenting complaint	Persistent dry, tickling cough for two weeks, worse at night
Medical history	Type 2 diabetes for 5 years. Newly diagnosed hypertension. Non-smoker.

Medication history

Medicine	Dose	Indication	Started
Metformin	500 mg twice daily	Type 2 diabetes	5 years ago
Enalapril	5 mg once daily	Hypertension	Day 1

Timeline

When	What happened
Day 1	Enalapril started.
Week 2	Dry cough begins. No fever, no sputum, no breathlessness.
Week 4	Cough continues. Chest examination and chest X-ray are normal.
Week 4	Enalapril stopped and replaced with losartan.
Week 6	Cough has completely resolved.

Analysis

Question	Finding
Temporal relationship	Yes. Cough started about one week after enalapril was begun.
Known reaction	Yes. Dry cough is a common, well-known effect of ACE inhibitors, linked to accumulation of bradykinin.
Alternative causes	Unlikely. No signs of infection, normal chest X-ray, non-smoker. Metformin has been taken for 5 years without cough.
Dechallenge	Positive. Cough resolved within two weeks of stopping enalapril.
Rechallenge	Not done.

Conclusion

Drug-related: Yes
Suspected drug: Enalapril (ACE inhibitor)
Causality (WHO-UMC): Probable / Likely
Type of reaction: Type A (related to the pharmacological action of the drug)
Seriousness: Non-serious
Action: Substitute with a drug from another class (an angiotensin receptor blocker) and document the reaction.

Case 3: Black stools in a patient taking several medicines

Patient	Male, 67 years
Presenting complaint	Black tarry stools for two days, weakness and tiredness
Medical history	Atrial fibrillation for 2 years. Osteoarthritis of the knee. No previous ulcer or bleeding.

Medication history

Medicine	Dose	Indication	Started
Warfarin	5 mg once daily	Atrial fibrillation	2 years ago (INR stable, in range)
Atenolol	50 mg once daily	Heart rate control	2 years ago
Diclofenac	50 mg twice daily	Knee pain	Day 1

Timeline

When	What happened
Day 1	Diclofenac started for knee pain.
Day 9	Patient notices black tarry stools.
Day 10	Admitted to hospital. Haemoglobin has fallen; INR is within the therapeutic range.
Day 10	Diclofenac stopped, warfarin withheld temporarily, proton pump inhibitor started.
Day 14	Bleeding has stopped. Patient discharged; warfarin restarted with monitoring.

Analysis

Question	Finding
Temporal relationship	Yes. Bleeding began 9 days after diclofenac was added.
Known reaction	Yes. Gastrointestinal bleeding is a known reaction to NSAIDs, and the risk is higher when taken with an anticoagulant.
Alternative causes	Warfarin can cause bleeding, but it had been taken for 2 years with a stable INR and no bleeding. Atenolol is not associated with GI bleeding. No previous ulcer disease.
Dechallenge	Positive. Bleeding stopped after diclofenac was withdrawn.
Rechallenge	Not done.

Conclusion

Drug-related: Yes
Suspected drug: Diclofenac (NSAID); interacting drug: warfarin (increases the bleeding risk)
Causality (WHO-UMC): Probable / Likely for diclofenac
Type of reaction: Type A
Seriousness: Serious (required hospitalisation)
Action: Avoid NSAIDs in this patient, use a safer analgesic, and report as a serious ADR.

Case 4: Fever and sore throat in a patient on long-term therapy

Patient	Female, 30 years
Presenting complaint	Fever, sore throat, body ache and runny nose for two days
Medical history	Hypothyroidism, well controlled. Two family members at home have had similar symptoms in the same week.

Medication history

Medicine	Dose	Indication	Started
Levothyroxine	50 mcg once daily	Hypothyroidism	2 years ago (no dose change)

Timeline

When	What happened
2 years ago	Levothyroxine started. Tolerated well, with no dose change since.
Day 1	Fever, sore throat and body ache begin, after family members fell ill.
Day 2	Patient asks whether her medicine could be the cause.
Day 6	All symptoms resolved. Levothyroxine was continued throughout.

Analysis

Question	Finding
Temporal relationship	No. The medicine has been taken for 2 years with no recent change.
Known reaction	No. Fever with sore throat and runny nose is not an expected reaction to levothyroxine at a stable dose.
Alternative causes	Yes. A viral upper respiratory infection is a clear explanation, supported by similar illness in the household.
Dechallenge	Not applicable. Symptoms resolved while the medicine was continued, which goes against a drug cause.
Rechallenge	Not applicable.

Conclusion

Drug-related: No
Suspected drug: None
Causality (WHO-UMC): Unlikely
Most likely cause: Viral upper respiratory tract infection
Action: Continue levothyroxine, treat the infection symptomatically, and reassure the patient.

Summary of findings

Case	Reaction	Drug-related?	Suspected drug	Causality
1	Skin rash, nausea, dizziness	Yes	Amoxicillin	Probable
2	Persistent dry cough	Yes	Enalapril	Probable
3	Gastrointestinal bleeding	Yes	Diclofenac (interaction with warfarin)	Probable
4	Fever and sore throat	No	None	Unlikely

Key learning points

- Timing is the first clue: a reaction that starts soon after a new medicine deserves suspicion.
- The newest medicine is usually the first suspect, but every medicine and the disease itself must be considered.
- Improvement after stopping the medicine (positive dechallenge) strongly supports a drug cause.
- Not every symptom in a patient on medication is an ADR; ruling out other causes is part of the assessment.
- Serious reactions, such as those needing hospitalisation, must be reported promptly.

Note: the case scenarios in this document are hypothetical and were written for learning purposes.